Bootstrapping Accountability and Privacy to IPv6 Internet without Starting from Scratch

Lin He, Gang Ren, Ying Liu✉
Institute for Network Sciences and Cyberspace, Tsinghua University, Beijing 100084, China

***Abstract*—Accountability and privacy are considered valuable but conflicting properties in the Internet, which at present does not provide native support for either. Past efforts to balance accountability and privacy in the Internet have unsatisfactory deployability due to the introduction of new communication identifiers, and because of large-scale modifications to fully deployed infrastructures and protocols.**

The IPv6 is being deployed around the world and this trend will accelerate. In this paper, we propose a private and accountable proposal based on IPv6 called PAVI that seeks to bootstrap accountability and privacy to the IPv6 Internet without introducing new communication identifiers and large-scale modifications to the deployed base. A dedicated quantitative analysis shows that the proposed PAVI achieves satisfactory levels of accountability and privacy. The results of evaluation of a PAVI prototype show that it incurs little performance overhead, and is widely deployable.

***Index Terms*—accountability, privacy, IPv6**

I. Introduction

Accountability and privacy are both considered valuable in the Internet, although they appear to conflict. However, the Internet does not at present provide native support for either. On the one hand, the lack of accountability makes it possible for attackers to spoof source addresses to launch massive distributed denial of service attacks. Network operators cannot stop ongoing attacks because they do not know the true senders and thus cannot effectively filter spoofing traffic. On the other hand, people have the right to anonymously access information through the Internet. However, the disclosure of pervasive monitoring and mass surveillance [1]–[4] causes people to become increasingly concerned about their privacy, and to seek applications and techniques [5], [6], though heavyweight, to protect it.

A considerable amount of research has been conducted on accountability and privacy in the last few decades. However, the mechanisms proposed in this context often ensure that one of the two properties is obtained at the cost of the other. The Accountable and Private Internet Protocol (APIP) [7], and the Accountable and Private Network Architecture (APNA) [8] are the main protocols that attempt to strike a balance between accountability and privacy at the network layer. However, both protocols have limitations (see more details in Section VIII), the most serious of which is that both introduce new identifiers to create communication connections that require large-scale modifications to existing Internet infrastructures and protocols.

✉ Ying Liu is the corresponding author (e-mail: liuying@cernet.edu.cn).

Under this circumstance, we ask the question: *"Can we bootstrap accountability and privacy to the current Internet without introducing new communication identifiers and large-scale modifications to fully deployed infrastructures and protocols?"* We are interested in this kind of approach because it can deliver practical benefits without having to build everything from scratch.

The IPv6 is being deployed increasingly commonly around the world, and this trend will accelerate. Approximately 23.5% of users access Google over IPv6 [9]. Belgium regularly delivers more than 50% of traffic to major content providers over IPv6 [10]. The IETF will stop requiring IPv4 compatibility in new or extended protocols, and its subsequent work will optimize for and depend on IPv6 [11].

In this paper, we propose a new private and accountable proposal based on IPv6 called PAVI that bootstraps accountability and privacy to the IPv6 Internet. PAVI takes full advantage of IPv6 addresses instead of creating new communication identifiers. Moreover, it uses standardized and well-maintained protocols with well-designed extensions as much as possible to simultaneously achieve accountability and privacy. These features allow PAVI to avoid large-scale modifications to infrastructures and protocols related to communication identifiers, and render it widely deployable.

Contributions. We make the following contributions:

- We propose PAVI that bootstraps accountability and privacy to the IPv6 Internet without introducing new communication identifiers and large-scale modifications to fully deployed infrastructures and protocols.
- We conduct a quantitative analysis to show that PAVI satisfies the demands of accountability.
- We validate the privacy goals of PAVI using information-theoretic entropy.
- We demonstrate the practicality and deployability of PAVI through the performance evaluation of a prototype. The results show that PAVI is a lightweight and deployable protocol.

The remainder of this paper is organized as follows: Section II presents the problem formulation, and Section III presents an overview of PAVI. Section IV describes the details of PAVI. In Section V, we detail a protocol analysis of PAVI, and describe its implementation and performance evaluation in Section VI. Related work is reviewed in Section VIII. Finally, we offer the conclusions of the paper in Section IX.

II. Problem Formulation

In this section, we describe the resources and protocols available, the design goals, and the adversary model.

A. What we have

The Internet has many resources and mechanisms that can be used to bootstrap accountability and privacy.

1) Large IPv6 Address Space: The address space of IPv6 is considerably larger than that of IPv4. For even a single /64 IPv6 prefix, the address space is 2^{32} times as large as the entire IPv4 address space. A host generally has two or three global unicast addresses, i.e., two configured via SLAAC [12] [1] and one configured via DHCPv6 [13], within a /64 IPv6 prefix. However, even if there are 1,000,000 hosts, the address space usage for the /64 prefix is only approximately $\frac{1}{2^{42}}$, a number little greater than zero. Of course, we do not take advantage of the entire IPv6 address space.

2) Multiple Addresses Per Host: As mentioned above, a host can generally obtain two SLAAC addresses and one DHCPv6 address within a prefix. One of the SLAAC addresses is used for outgoing connections and the other for incoming connections. However, even if we assume that every host is assigned 100 addresses, the space usage for a prefix is still less than $\frac{1}{2^{37}}$. In short, IPv6 address usage practices remind us that hosts can be assigned more addresses to achieve other goals, e.g., privacy protection.

3) Standardized and Well-maintained Protocols: For a long time, protocols related to address allocation and security have been created and maintained by the IETF, and include DHCPv6 [13], SAVI [14], IPsec [15], and RPKI [16]. Using these protocols instead of creating new, similar ones can save us a lot of effort. Moreover, while creating a new protocol, we often encounter problems similar to those faced when working with existing protocols. For example, APNA [8] encounters the same certificate management issue, as does IPsec.

B. Design Goals

1) Accountability: Accountability in general means being held responsible for one's actions. At the network layer in computing, accountability refers to the characteristic whereby all data packets can be attributed to their senders. Achieving accountability in the Internet involves a number of procedures. First, the authenticity of data packets should be ensured because they are used in the network to account for misbehaving senders. Second, each data packet should be associated with a specific sender identifier. Third, every sender should have a unique identifier in the network. Many useful protocols can also be designed to use accountability to achieve specific goals, e.g., blocking and filtering traffic, and scoring each sender.

2) Privacy: As shown in Table I, our goals with respect to privacy are four. The first is sender anonymity, which provides the capability for a sender to hide its identity from third-party observers in the source AS, transit ASes, destination AS, and even the receiver. We do not consider sender anonymity to be against observers in the LAN segment because they already know the identity (link layer address) of the sender. Nor do we consider sender anonymity to be against the source AS because it knows the identity and network attachment of the sender. Our second privacy goal is sender-flow unlinkability, which means that an adversary cannot obtain more information about the source relativity of flows by observing any number of flows originating from the same AS. The source relativity of flows refers to the possibility of two flows originating from the same sender. The third goal is sender-receiver unlinkability. No observer can obtain more information about whether the sender is communicating with the same receiver by observing flows. Note that for those who know the identity of one of the sender and the receiver, achieving sender-receiver unlinkability is to ensure the privacy of the identity of the other. The fourth goal of privacy here is data confidentiality. Only the receiver can know the packet payload, and all other entities, including the source AS, should know nothing about it. In this paper, we delegate the privacy of packet payload to well-established protocols, e.g., IPsec [15]. Note that privacy leakage from higher layers is beyond the scope of this paper.

3) Deployability: Compared with prevalent protocols [7], [8], we aim to make PAVI more deployable, which implies two sub-goals:

Lightweight enhancements: PAVI introduces only lightweight enhancements to the Internet, and does not require new communication identifiers and large-scale modifications to fully deployed infrastructures and protocols, unlike [7], [8]. PAVI only adds a slight performance overhead to the deployed Internet base.

Incrementally deployable: PAVI is incrementally deployable. Upgraded ASes can benefit from a partially deployed PAVI immediately whereas legacy ASes can continue operating.

C. Adversary Model

The goal of the adversary against the accountability of PAVI is to create packets associated with someone else in the network without being identified, whereas the goal against the privacy of PAVI is to determine the identity of a sender, whether two flows from the same source AS originate from the same sender, whether two flows have the same sender and receiver identities, or the packet payload. We assume that the adversary can observe packet headers and payload. It is assumed that the hosts can be compromised and used to launch attacks. Note that secret keys used in PAVI are well protected, and the adversary cannot compromise them.

III. Design Overview

In this section, we describe a high-level overview of PAVI. The protocol details are provided in Section IV. We introduce the notion of accountable privacy addresses (APAs) that can be used to associate with users in a cryptographic and stateless manner. In essence, APAs are short-lived IPv6 addresses with user identities embedded. They can be used to account for misbehaving users and protect user privacy simultaneously.

[1] Although this is not mentioned in [12], it is common practice for most current operation systems to configure two SLAAC addresses for hosts.

TABLE I
PRIVACY GOALS IN PAVI.

Observers	Sender anonymity	Sender-flow unlinkability	Sender-receiver unlinkability	Data confidentiality
Observers in source LAN segment	×	×	√	√
Observers in source AS	√	√	√	√
Source AS	×	×	√	√
Transit ASes	√	√	√	√
Destination AS	√	√	√	√
Observers in destination AS	√	√	√	√
Observers in destination LAN segment	√	√	√	√
Receiver	√	√	√	×

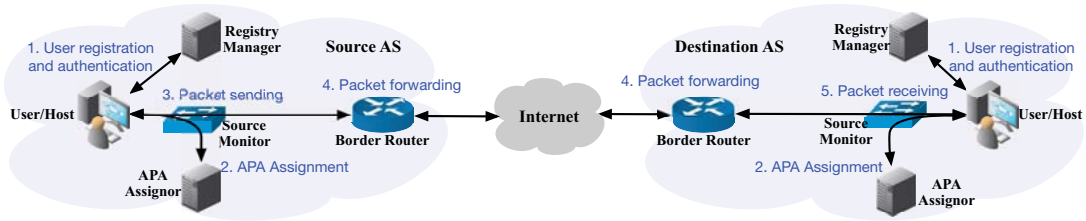


Fig. 1. High-level overview of PAVI

In each AS, many entities join the communication.

- **Users** use hosts to access the Internet.
- **Hosts** send or receive packets to or from the Internet.
- **Registry Manager** authenticates hosts and shares secret symmetric keys with hosts.
- **APA Assigner** generates APAs and assigns them to hosts.
- **Source Monitor** observes address assignment procedures, creates binding entries, and filters spoofing traffic.
- **Border Router** masks the source and destination addresses of packets and forwards packet flows.

As shown in Fig. 1, the communication between hosts consists of five steps:

1) **User Registration and Authentication:** A user subscribes to a preferred privacy service at its AS, authenticates to its AS, and shares a symmetric key with it securely during authentication (see Section IV-B).
2) **APA Assignment:** The host in which the user resides sends APA requests to the APA Assigner and configures APAs embedded with his/her UID (see Section IV-C).
3) **Packet Sending:** The source host sends packets each with a randomly chosen APA, integrity protection over the packet, and a masked destination address (see Section IV-D).
4) **Packet Forwarding:** The border routers of the source and destination AS mask the destination and source addresses, respectively, to increase sender-receiver unlinkability (see Section IV-E).
5) **Packet Receiving:** The destination host unmasks the source address of each packet to communicate with the source host (see Section IV-F).

IV. DETAILS OF PAVI

This section presents details of the design of PAVI. Table II summarizes the notation we use while describing the protocol.

TABLE II
NOTATION.

Symbol	Remarks
$\mathcal{H}_{\mathcal{S}}^{\mathcal{U}_i}$	The source host used by user $\mathcal{U}_i$
$\mathcal{H}_{\mathcal{D}}^{\mathcal{U}_j}$	The destination host used by user $\mathcal{U}_j$
$\mathsf{A}_{\mathcal{S}}$	The original source address
$\mathsf{A}_{\mathcal{D}}$	The original destination address
UID_i	User identifier (UID) assigned to user $\mathcal{U}_i$
N_i	The maximum number of APAs assigned to user $\mathcal{U}_i$
$k_{\mathcal{A}_i}$	Secret key of AS $\mathcal{A}_i$
$k_{\mathcal{A}_i\mathcal{H}_j}$	Secret key shared between AS $\mathcal{A}_i$ and host $\mathcal{H}_j$
$k_{\mathcal{A}_i\mathcal{A}_j}$	Secret key shared between AS $\mathcal{A}_i$ and AS $\mathcal{A}_j$
$E_k(\cdot)$	Symmetric encryption using key k
$E_k^{-1}(\cdot)$	Symmetric decryption using key k
$MAC_k(\cdot)$	Message authentication code (MAC) using key k

A. Assumptions

- *All entities can verify prefixes to which an AS can originate routes.* For example, a scheme such as RPKI [16] can be used by legitimate holders to authorize ASes to originate routes to one or more prefixes.
- *Each AS shares secret symmetric keys with other ASes.* After obtaining public keys from the RPKI, the ASes can use a non-interactive Diffie-Hellman key exchange [17], [18] to establish secret symmetric keys.
- *We assume that the cryptographic mechanisms are secure.* That is, encryptions cannot be violated and MACs cannot be forged.

B. User Registration and Authentication

PAVI aims to provide privacy to users. Internet service providers (ISPs) have business incentives to provide privacy to customers. In PAVI, ISPs are allowed to provide different levels of privacy to customers. It is assumed that an AS has M users $\{\mathcal{U}_1, \mathcal{U}_2, \cdots, \mathcal{U}_M\}$. When user $\mathcal{U}_i$ subscribes to an ISP, the ISP creates a unique user identifier UID_i and the corresponding authentication credentials for him/her. $\mathcal{U}_i$ can also purchase different levels of privacy by choosing the preferred number N_i of APAs that the ISP assigns to the user.

As PAVI associates users with packets by embedding UIDs into APAs, it is necessary to perform user authentication before APA assignment. The Registry Manager uses a layer-2 authentication mechanism to authenticate users. Several well-designed authentication mechanisms can be used, e.g., the RADIUS protocol [19] and the Diameter base protocol [20]. During authentication, the host creates a secret symmetric key $k_{\mathcal{A}_S\mathcal{H}_S}$ and securely shares it with the Registry Manager, which maintains an authenticated user list and keeps updating it with other entities in the AS, including the APA Assigner and border routers. The Registry Manager also sends (UID_i, $k_{\mathcal{A}_S\mathcal{H}_S}$) to the border routers.

C. APA Assignment

The goal of the APA assignment procedure is to assign N_i APAs embedded with UID_i to the requesting host $\mathcal{H}_S^{\mathcal{U}_i}$. Following user authentication, $\mathcal{H}_S^{\mathcal{U}_i}$ begins to request APAs from the APA Assigner. As the Registry Manager has synchronized the update with the authenticated user list, the APA Assigner generates N_i APAs $\{\mathsf{A}_1^{\mathcal{U}_i}, \mathsf{A}_2^{\mathcal{U}_i}, \cdots, \mathsf{A}_{N_i}^{\mathcal{U}_i}\}$ and assigns them to $\mathcal{H}_S^{\mathcal{U}_i}$. Each APA $\mathsf{A}_j^{\mathcal{U}_i}$ is generated from

$$\mathsf{A}_j^{\mathcal{U}_i} = prefix||E_{k_{\mathcal{A}_S}}(UID_i||n_j||t_j||d_j), \tag{1}$$

where $prefix$ is the prefix used to generate $\mathsf{A}_j^{\mathcal{U}_i}$, n_j is a counter employed to indicate the number of $\mathsf{A}_j^{\mathcal{U}_i}$, t_j is the time at which the APA Assigner generates $\mathsf{A}_j^{\mathcal{U}_i}$, d_j is a counter employed to resolve duplicate detection and incremented by 1 for each new generation, and $||$ represents concatenation. Note that these APAs may have different preferred lifetimes. A source monitor observes the APA assignment procedure and creates binding entries between APAs and trust anchors (e.g., switch port). Such mechanisms as [14] can be used to achieve this goal.

D. Packet Sending

Once a source host $\mathcal{H}_S^{\mathcal{U}_i}$ obtains $\mathsf{A}^{\mathcal{U}_i} = \{\mathsf{A}_1^{\mathcal{U}_i}, \mathsf{A}_2^{\mathcal{U}_i}, \cdots, \mathsf{A}_{N_i}^{\mathcal{U}_i}\}$ from its AS, it can start communicating with other hosts. If $\mathcal{H}_S^{\mathcal{U}_i}$ decides to protect data confidentiality, it uses the ESP header [21]. Each packet flow originating from $\mathcal{H}_S^{\mathcal{U}_i}$ is processed in the following three steps:

1) APA Determination: Once the peer host has been determined (e.g., through DNS lookup), $\mathcal{H}_S^{\mathcal{U}_i}$ sends packet flows each with a source APA randomly chosen from $\mathsf{A}^{\mathcal{U}_i}$. Sending flows using different source APAs improves unlinkability between a sender and its flows. The number of packet flows using $\mathsf{A}_j^{\mathcal{U}_i}$ depends on $\mathcal{H}_S^{\mathcal{U}_i}$, and one flow per APA is the best.

2) PAVI Header Insertion: $\mathcal{H}_S^{\mathcal{U}_i}$ must insert the PAVI header into each packet to protect the integrity of the packets. Details of the PAVI header are as follows:

- Next Header: a 1-byte field that identifies the type of the next payload following the PAVI header.
- Payload Length: a 2-byte field that specifies the length of the PAVI header in 4-byte units.
- Reserved: a 1-byte field reserved for future use. It must be set to zero by the sender and ignored by the receiver.
- Extra Encrypted Address (EEA): an 8-byte value used by hosts and border routers to store extra encrypted address.

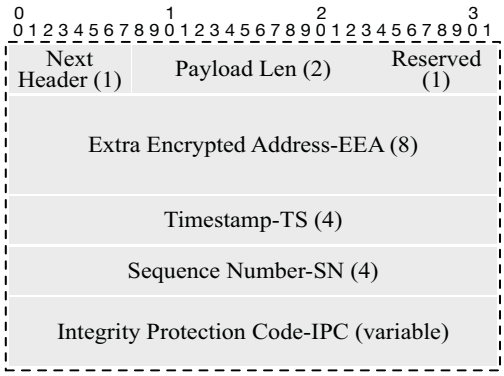


Fig. 2. PAVI header

- Timestamp (TS): a 4-byte value that identifies the time at which the sender inserts the PAVI header into the packet.
- Sequence Number (SN): a 4-byte field that contains a counter that increases by one for each packet.
- Integrity Protection Code (IPC): a variable-length field that contains the integrity check value for the packet. Note that this value must be an integral multiple of four bytes in length. The IPC is generated from

$$\begin{aligned} IPC =& MAC_{k_{\mathcal{A}_S\mathcal{H}_S}}(IP_{src}||IP_{dst}||EEA||TS \\ &||SN||H(payload)), \end{aligned} \tag{2}$$

where $H(payload)$ is the hash encryption of the payload.

As the IPC is computed over the packet using the secret symmetric key $k_{\mathcal{A}_S\mathcal{H}_S}$ shared with its AS, this means that the PAVI header is used to improve the accountability of the packet instead of it being verified by the peer host.

3) Destination Masking: $\mathcal{H}_S^{\mathcal{U}_i}$ encrypts the interface identifier (IID) of the destination IPv6 address of each packet using the secret symmetric key $k_{\mathcal{A}_S\mathcal{H}_S}$ as follows:

$$IID_{\mathcal{A}_S\mathcal{H}_S}||EEA_{\mathcal{A}_S\mathcal{H}_S}=E_{k_{\mathcal{A}_S\mathcal{H}_S}}(IID_{\mathcal{D}}||TS||SN). \tag{3}$$

Then, $\mathcal{H}_S^{\mathcal{U}_i}$ uses the first 64 bits of the result of encryption as the new IID of the destination address and the last 64 bits as EEA in the PAVI header, and sends the packets. The goal of this step is to improve sender-receiver unlinkability against observers in the same LAN segment as $\mathcal{H}_S^{\mathcal{U}_i}$.

E. Packet Forwarding

When a border router of the source AS receives an outgoing packet, it first verifies its integrity by computing an IPC using the IP header and payload, and comparing it with the one in the PAVI header. If the validation fails, the border router drops the packet. Otherwise, it uses the secret key $k_{\mathcal{A}_S\mathcal{H}_S}$ to decrypt the IID of the destination address and EEA in the PAVI header to obtain the original destination address $\mathsf{A}_{\mathcal{D}}$. Then, it re-encrypts the IID of $\mathsf{A}_{\mathcal{D}}$ using secret key $k_{\mathcal{A}_S\mathcal{A}_D}$ shared with the destination AS and forwards the packet.

$$\begin{aligned} IID_{\mathcal{A}_S\mathcal{A}_D}||EEA_{\mathcal{A}_S\mathcal{A}_D} =& E_{k_{\mathcal{A}_S\mathcal{A}_D}}(E^{-1}_{k_{\mathcal{A}_S\mathcal{H}_S}}(IID_{\mathcal{A}_S\mathcal{H}_S} \\ &||EEA_{\mathcal{A}_S\mathcal{H}_S})). \end{aligned} \tag{4}$$

In summary, only authenticated packets can leave the source AS following these checks.

When a border router of the destination AS receives an incoming packet, it first uses the source prefix to search the

secret symmetric key $k_{\mathcal{A}_S\mathcal{A}_D}$ shared with the source AS. It then decrypts the IID of the destination address and EEA in the PAVI header to obtain the original destination address $\mathsf{A}_{\mathcal{D}}$.

$$IID_{\mathcal{D}}||TS||SN = E^{-1}_{k_{\mathcal{A}_S\mathcal{A}_D}}(IID_{\mathcal{A}_S\mathcal{A}_D}||EEA_{\mathcal{A}_S\mathcal{A}_D}). \quad (5)$$

After encrypting the IID of the original source address $\mathsf{A}_{\mathcal{S}}$ with secret symmetric key $k_{\mathcal{A}_D\mathcal{H}_D}$ shared with the destination host $\mathcal{H}^{\mathcal{U}_i}_{\mathcal{D}}$, it forwards the packet. Note that if the sender uses each APA to send only one flow, the encryption of the original source address can be ignored.

$$IID_{\mathcal{A}_D\mathcal{H}_D}||EEA_{\mathcal{A}_D\mathcal{H}_D} = E_{k_{\mathcal{A}_D\mathcal{H}_D}}(IID_{\mathcal{S}}||TS||SN). \quad (6)$$

F. Packet Receiving

When $\mathcal{H}^{\mathcal{U}_i}_{\mathcal{D}}$ receives the packet, it decrypts the IID of the source address to determine the real source address $\mathsf{A}_{\mathcal{S}}$ of the host with which it communicates:

$$IID_{\mathcal{S}}||TS||SN = E^{-1}_{k_{\mathcal{A}_D\mathcal{H}_D}}(IID_{\mathcal{A}_D\mathcal{H}_D}||EEA_{\mathcal{A}_D\mathcal{H}_D}). \quad (7)$$

Once $\mathcal{H}^{\mathcal{U}_i}_{\mathcal{D}}$ assumes that packet flows from the sender are malicious or unwanted, $\mathcal{H}^{\mathcal{U}_i}_{\mathcal{D}}$ can require its AS to block traffic. Once $\mathcal{A}_D$ receives the block request, it sends the request to $\mathcal{A}_S$ with the signature signed by its private key. $\mathcal{A}_S$ blocks traffic from the specific APA if the block request is valid, such as through Shutoff [7], [8].

V. Protocol Analysis

In this section, we show how PAVI prevents attacks that undermine accountability and privacy, and quantitatively analyze accountability and privacy.

A. Security Analysis

1) Spoofing APAs: The adversary can attempt to generate APAs by itself. However, packets with spoofing APAs are dropped because the source monitor validates the source address of each packet according to its binding entries.

2) Integrity Attacks: A malicious host may send packets with incorrect IPCs. In PAVI, the border router checks the IPCs to determine whether to forward the packets. If the IPC check fails, the border router drops the packets.

3) Identity Minting: Identity minting is when a malicious user attempts to create and use unauthorized UIDs to connect to the Internet. However, PAVI allows an AS to provide only a unique UID to every user within the AS. Unauthorized UIDs fail the user authentication process.

4) Data Confidentiality Breach: Adversaries may attempt to eavesdrop on data packets and decrypt the payload. However, they cannot succeed in decryption without secret keys. The security of ESP depends on both the strength of the cryptographic algorithms chosen and that of the keys used with them. Requirements for the implementation of the cryptographic algorithm, and usage guidance for ESP and AH are provided in [22].

B. Quantitative Analysis

1) Accountability Analysis: We choose an AS as a unit to discuss accountability. The goal of accountability at the network layer is to determine the true user identities of all the data packets originating from an AS. The accountability of an AS $\mathscr{S}$ can be defined as:

$$A_{\mathscr{S}}(P) = \frac{\sum_{p \in P} a(p)}{||P||}, \quad (8)$$

where P is a set containing all data packets originating from $\mathscr{S}$, p is a packet in P, $||P||$ is the total number of packets in P, and $a(p)$ is the accountability estimation for p. Note that $a(p)$ is a value between zero and one.

Before discussing how to compute $a(p)$, we first present several related definitions. $t(p)$, a Boolean value, is the authenticity of packet p. Specifically, the packet is "true" when $t(p) = 1$ and "not true" when $t(p) = 0$. $r(p)$ is used to extract the sender identifier from p. $u_T(id)$ is the number of entities that use id during period T. Then, $a(p)$ is defined as follows:

$$a(p) = \frac{t(p)}{u_T(r(p))}. \quad (9)$$

TABLE III
Comparison of accountability.

	$t(p)$	$r(p)$	$u_T(id)$	$a(p)$	$A_{\mathscr{S}}(P)$
Today's Internet	0	IP	N/A	0	0
NAT (True source)	1	IP	N	$\frac{1}{N}$	$\frac{1}{N}$
Persona	1	IP	M	$\frac{1}{M}$	$\frac{1}{M}$
APIP	1/0	Public key	1	1/0	$\frac{\|\|V\|\|}{\|\|P\|\|}$
APNA	1	HID	1	1	1
PAVI	1	UID	1	1	1

We then compute the accountability of several protocols under the worst-case scenario, as shown in Table III. N denotes the number of hosts mapping to the same IP address during period T using NAT, and M denotes the number of hosts that use the same IP address during period T in Persona. For APIP, only the first packet is authentic and vouched for by the host's accountability delegate because "whitelisted" hosts can ignore sending packet fingerprints to their delegates. Therefore, only a few packets can be accounted for during the verification cycle. V is the collection of the first packets verified along the forwarding path during each verification cycle. As the APNA embeds host identifiers into their newly defined addresses (i.e., EphIDs) and uses MACs to protect the authenticity and integrity of packets, it has good accountability. PAVI adopts well-designed monitoring techniques to prevent source spoofing, uses IPCs to protect the integrity of packets, and associates users with their packets via APAs, which also yields satisfactory accountability.

2) Privacy Analysis: We use information-theoretic entropy [23] as the measure of information contained in a probability distribution to analyze sender anonymity, sender-flow unlinkability, and sender-receiver unlinkability. Recall that there are M users $\mathcal{U} = \{\mathcal{U}_1, \mathcal{U}_2, \cdots, \mathcal{U}_M\}$ in an AS. $\mathcal{U}_i$ has N_i APAs $\mathsf{A}^{\mathcal{U}_i} = \{\mathsf{A}^{\mathcal{U}_i}_1, \mathsf{A}^{\mathcal{U}_i}_2, \cdots, \mathsf{A}^{\mathcal{U}_i}_{N_i}\}$ and uses each APA to send flows. Therefore, we have $\mathsf{A} = \{\mathsf{A}^{\mathcal{U}_i}_j | 1 \le i \le M, 1 \le j \le N_i\}$.

Sender Anonymity. We measure the possible information leakage of sender identity based on degree of anonymity [24]. For example, an adversary $\mathscr{T}$ detects many APAs by observing traffic originating from an AS. The goal of the adversary is to exploit the user identity of each APA. For each

APA, it maintains a probability distribution X. Specifically, the adversary $\mathscr{T}$ believes that it belongs to user $\mathcal{U}_i$ with probability $p_i := P_{\mathscr{T}}(X = \mathcal{U}_i)$. Then, the entropy is defined as:

$$H_{\mathscr{T}}(X) = -\sum_{i=1}^{M} p_i \cdot log_2(p_i). \tag{10}$$

The maximum entropy is $max(H_{\mathscr{T}}(X))$=$log_2(M)$ when p_i=$\frac{1}{M}$. As $\mathscr{T}$ does not know the secret key of the AS, it cannot decrypt the APA to obtain the user identifier. Accordingly, the adversary believes that the APA can belong to any user in the AS with probability p_i=$\frac{1}{M}$. Therefore, we can calculate entropy $H_{\mathscr{T}} = -\sum_{i=1}^{M} \frac{1}{M} \cdot log_2(\frac{1}{M}) = log_2(M)$. Based on [24], the degree of anonymity after introducing PAVI is:

$$d = 1 - \frac{max(H_{\mathscr{T}}(X)) - H_{\mathscr{T}}}{max(H_{\mathscr{T}}(X))} = \frac{H_{\mathscr{T}}}{max(H_{\mathscr{T}}(X))} = 1. \tag{11}$$

Although each APA contains a user identifier, the adversary cannot obtain the real user identifier from it due to encryption protection. However, the IPv6 Internet uses IPv6 addresses to identify hosts for quite a long time, which makes it easy for $\mathscr{T}$ to determine the host's identity and obtain $d = 0$.

Sender-Flow Unlinkability. We measure sender-flow unlinkability based on the degree of unlinkability [25]. An adversary $\mathscr{T}$ observes outgoing traffic originating from the AS and detects $N = \sum_{i=1}^{M} N_i$ different source APAs. If the adversary knows nothing about users in the AS, the unlinkability of the APAs in A for $\mathcal{U}$ against the adversary $\mathscr{T}$ is defined as:

$$U_{\mathsf{A}}(\mathscr{T}_{\emptyset}, \mathcal{U}) = log_2(B_N), \tag{12}$$

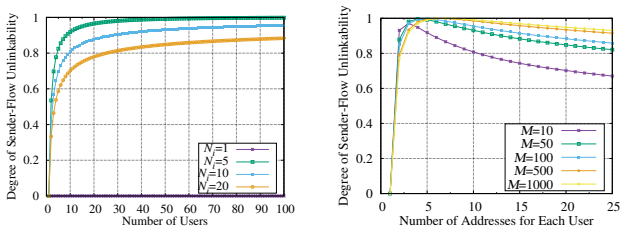
where $\mathscr{T}_{\emptyset}$ denotes the adversary without any hint and B_N is the total number of partitions of A, a Bell number [26]. However, if $\mathscr{T}$ knows a hint H whereby M users in the AS send flows, the goal of the adversary is to partition set A into M non-empty subsets. Then, the unlinkability of the APAs in A for $\mathcal{U}$ against adversary $\mathscr{T}$ is $U_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U}) = log_2(S(N, M))$, where $\mathscr{T}_{\mathsf{H}}$ denotes that the adversary with hint H and $S(N, M)$ is a Stirling number of the second kind [27]. Therefore, the degree of unlinkability of the APAs in A for $\mathcal{U}$ against adversary $\mathscr{T}$ with access to hint H is:

$$D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U}) = \frac{U_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})}{U_{\mathsf{A}}(\mathscr{T}_{\emptyset}, \mathcal{U})} = \frac{log_2(S(N, M))}{log_2(B_N)}. \tag{13}$$

Fig. 3 shows the variation in the degree of sender-flow unlinkability $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})$ with respect to (a) the number of users M and (b) the number of addresses N_i assigned to each user in an AS. We see that $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})$ increases with M. However, as N_i increases, $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U})$ first increases quickly and then decreases slightly. An AS can determine the minimum degree of sender-flow unlinkability, and we suggest that an AS should provide degree $D_{\mathsf{A}}(\mathscr{T}_{\mathsf{H}}, \mathcal{U}) \geq 0.8$. Nevertheless, the minimum acceptable degree for an AS may depend on the unlinkability requirement for the AS, and we believe that such a minimum cannot be suggested before tests.

Choose How Many APAs? Assume a new user U_{M+1} subscribes to N_{M+1} APAs. Then the degree of unlinkability of his APAs in $\mathsf{A}' = \{\mathsf{A}_j^{\mathcal{U}_i} | 1 \leq i \leq M+1, 1 \leq j \leq N_i\}$ against the adversary $\mathscr{T}$ can be defined as:

$$D_{\mathsf{A}'}(\mathscr{T}_{\emptyset}, U_{M+1}) = 1 - \frac{log_2(B_N)}{log_2(B_{N+N_{M+1}})}. \tag{14}$$



(a) (b)

Fig. 3. The variation in the degree of sender-flow unlinkability with respect to the number of users and addresses assigned to each user in an AS.

Obviously, the more APAs U_{M+1} subscribes to, the higher unlinkability he obtains. However, more APAs means more cost. Therefore, users should take unlinkability and cost into consideration when subscribing to APAs.

Sender-Receiver Unlinkability. Similar to the measurement of sender-flow unlinkability, we can also measure sender-receiver unlinkability based on the degree of unlinkability. The hints related to the sender, receiver, and their ASes may help the adversary make decisions. For example, an adversary may compromise hosts residing in the same LAN segment as the sender and obtain hint H_S concerning sender identity that leads to a decline in sender-receiver unlinkability. When a sender sends packets, compromised hosts in the LAN segment can learn the source and destination addresses. In PAVI, however, as the sender encrypts the IID of the destination address in each packet, compromised hosts cannot know the true destination address. The adversary can only guess one of the addresses within the destination prefix with probability $p_i = \frac{1}{2^{64}}$. The degree of unlinkability between sender and receiver against $\mathscr{T}$ with access to hint H_S is defined as:

$$D(\mathscr{T}_{\mathsf{H}_S}) = \frac{U(\mathscr{T}_{\mathsf{H}_S})}{U(\mathscr{T}_{\emptyset})} = \frac{-\sum_{i=1}^{2^{64}} \frac{1}{2^{64}} \cdot log_2(\frac{1}{2^{64}})}{-\sum_{i=1}^{2^{128}} \frac{1}{2^{128}} \cdot log_2(\frac{1}{2^{128}})} = \frac{1}{2}. \tag{15}$$

Moreover, if the adversary $\mathscr{T}$ also knows hint H_R whereby L users are in the destination AS $\mathcal{A}_{\mathcal{D}}$, the degree of unlinkability decreases to:

$$D(\mathscr{T}_{\mathsf{H}_S\mathsf{H}_R}) = \frac{U(\mathscr{T}_{\mathsf{H}_S\mathsf{H}_R})}{U(\mathscr{T}_{\emptyset})} = \frac{-\sum_{i=1}^{L} \frac{1}{L} \cdot log_2(\frac{1}{L})}{-\sum_{i=1}^{2^{128}} \frac{1}{2^{128}} \cdot log_2(\frac{1}{2^{128}})} = \frac{log_2 L}{128}. \tag{16}$$

According to the knowledge of the adversary, we calculate the degree of sender-receiver unlinkability in different scenarios. The results are shown in Table IV. The first row refers to knowledge of the receiver and its AS, and the first column indicates knowledge of the sender and its AS. We consider three typical hints concerning the sender/receiver:

- $\mathsf{H}_S^1/\mathsf{H}_R^1$: the adversary knows nothing about the sender/receiver and its AS.
- $\mathsf{H}_S^2/\mathsf{H}_R^2$: the adversary knows the number (M/L) of users in $\mathcal{A}_{\mathcal{S}}/\mathcal{A}_{\mathcal{D}}$.
- $\mathsf{H}_S^3/\mathsf{H}_R^3$: the adversary knows the sender/receiver identity.

VI. Implementation and Evaluation

A. Implementation

For source monitors, we used H3C E552 switches that implemented source address validation improvement (SAVI)

TABLE IV
COMPARISON OF SENDER-RECEIVER UNLINKABILITY.

	H_R^1	H_R^2	H_R^3
H_S^1	1	$\frac{64+log_2 L}{128}$	$\frac{1}{2}$
H_S^2	$\frac{64+log_2 M}{128}$	$\frac{log_2 M+log_2 L}{128}$	$\frac{log_2 M}{128}$
H_S^3	$\frac{1}{2}$	$\frac{log_2 L}{128}$	0

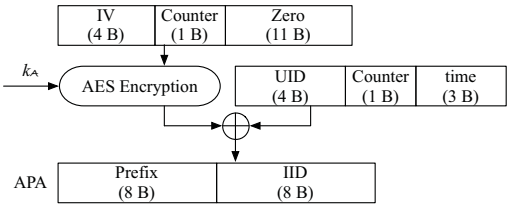


Fig. 4. APA generation procedure.

[14] to create binding entries and filter source spoofing traffic.

APA Generation: We obtained a concatenation of a 32-bit UID, an 8-bit indicating counter, a 20-bit timestamp, and a 4-bit detecting counter. Then, according to different encryption algorithms, the procedures differ. If we adopt a 64-bit block cipher, we need only use the secret key k_{A_S} to encrypt the concatenation and append the result to the prefix to obtain an APA. If a 128-bit block cipher is used, we can use the cipher to generate encryption keys, and then use the keys to encrypt the concatenation as shown in Fig. 4.

APA Assignment Implementation: We implemented the authentication process using 802.1x and the Radius protocol [19]. We used DHCPv6 servers and APA generators to implement the APA Assigner. APA generators aim to use UIDs to generate APAs after authentication. Fig. 5 sheds light on details of the APA assignment procedure. We used an Ubuntu server installed with the open-source ISC DHCP server software [28] as the DHCPv6 server, in which we modified the processing logic for the identity association for temporary addresses (IA_TA) options. A DHCPv6 server assigned APAs in IA_TA to hosts once it obtained APAs from the APA generators.

Border Router Implementation: We implemented the functionalities of the border routers based on the Intel Data Plane Development Kit [29]. To mask the source and destination addresses, we used the AES algorithm through Intel's AES-NI encryption instruction set [30]. To compute the IPC, we leveraged the cipher block chaining mode with AES as the underlying block cipher.

B. Evaluation

1) APA Generation: The APA generation overhead is inherent in cryptography-based security mechanisms. For comparison, we list the APA generation overheads of well-known 64-bit and 128-bit block ciphers in Table V. We used each algorithm to generate 1,000,000 APAs and calculated the average time. The tests were conducted using *OpenSSL* on a virtual machine with Intel Core i7-6700 CPU (one core, 3.4 GHz) and 4 GB of memory.

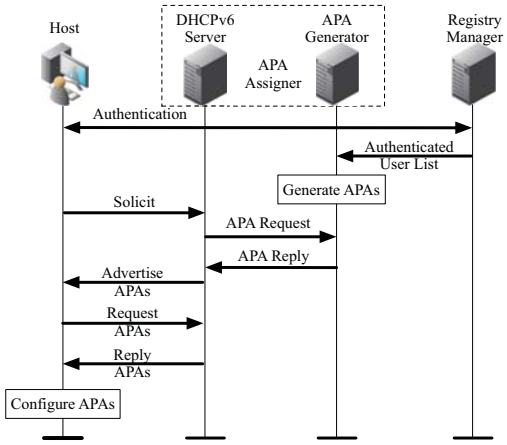


Fig. 5. Implementation of address assignment.

TABLE V
SECURITY LEVEL, ENCRYPTION SIZE, AND COMPUTATIONAL COSTS OF WELL-KNOWN ENCRYPTION SCHEMES.

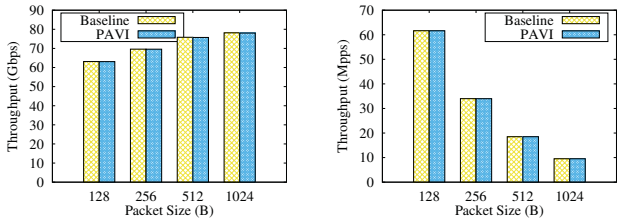
Scheme	Security	Block Size	Time
DES	56 bits	64 bits	109 ns
3DES	168 bits	64 bits	270 ns
Blowfish	256 bits	64 bits	66 ns
IDEA	128 bits	64 bits	83 ns
CAST-128	128 bits	64 bits	74 ns
SEED	128 bits	128 bits	99 ns
Camellia-128	128 bits	128 bits	51 ns
AES-128	128 bits	128 bits	15 ns

Storage Overhead: Consider the DHCPv6 client on Ubuntu, for example. A DHCPv6 lease with one APA in the IA_TA requires about 480 B of storage, including interface, options, and APA information. If we assume that a host is assigned 100 APAs, the storage overhead is about 12 KB. For the APA Assigner, if we assume that an AS has 1,000,000 hosts, it requires about 11.2 GB of disk space to store the APAs when APA Assigner assigns 100 APAs to each host.

2) APA Assignment Overhead: As we implemented the functionality of APA assignment based on IA_TA in DHCPv6, we evaluated the performance of the APA Assigner's response to a message from a host. When receiving a Solicit message, the APA Assigner used about 612 μs to build the Advertise message to respond to the host. When receiving a Request message, the APA Assigner used about 526 μs to build the Reply message for the host.

3) Data Forwarding: As a border router checks the IPC of the PAVI header and masks the source and destination addresses, we evaluated its forwarding performance. We used two Dell commodity servers with two 2.40 GHz Intel Xeon E5-2620 v3 CPUs, 16 GB of memory, and four dual-port 10 Gb NICs (a total of 80 Gbps). One server worked as a border router and directly connected to the other packet generator server installed with MoonGen [31].

Bandwidth Overhead: We tested the forwarding performance of all ports for four packet sizes: 128, 256, 512, and 1024 bytes. Fig. 6 shows the results. The baseline for the experi-



(a) Bit rate (b) Packet rate

Fig. 6. Verification at border routers

ments was the forwarding performance of legacy IPv6 traffic. The PAVI header processing throughput at the border routers matched well with the baseline because it only involved one MAC computation, two encryptions, and two lookups. If we assume that the average packet size was 540 bytes [32], sending packets with the PAVI header consumed an additional 6.25% of the bandwidth of the original traffic.

Storage Overhead: The storage overhead has two aspects: shared keys with ASes and shared keys with hosts. We assumed that the length of the keys shared with ASes was 128 bits (AES-128) and the length of key index was 4 bytes; then, the border router used 1.25 MB to store keys shared with 65,536 ASes. For keys shared with hosts, if we assume that 1,000,000 users in the AS, the storage cost was about 19 MB.

4) Comparison with Other Protocols: We compared PAVI with APIP and APNA shown in Table VI. PAVI required fewer modifications to the Internet architecture and infrastructures because it is built on IPv6. As it does not introduce new communication identifiers, modifications to infrastructures and protocols related to communication identifiers were not required, including applications, routing protocols, and DNS records. However, APIP and APAN required these modifications. Note that specific gateways for these protocols can be used to eliminate the modifications to hosts.

VII. Discussion

Block Cipher Length: In PAVI, we can use either 64-bit or 128-bit block ciphers to generate APAs. The difference concerns whether to store extra information used to generate APAs. For example, when using the AES algorithm, an AS should store the IV for each APA to obtain UIDs from each APA thereafter. As each user subscribes to the privacy service, it is necessary for an AS to store IV values of each user. But if the AS chooses IDEA algorithm, it can use secret keys to decrypt APAs to obtain UIDs directly without storing IV values. However, the advantage of AES is that it is often accelerated by hardware [30].

Partial Deployment: Regardless of whether transit ASes deploy PAVI, they should forward packets with the PAVI header. If $\mathcal{A}_D$ does not deploy PAVI, it should at lease support the decryption of the destination IID and EEA in the PAVI header. If $\mathcal{A}_D$ does not support destination IID decryption, source border routers do not mask the destination address of each packet.

Source and Destination Masking: The sender's masking destination addresses, the source border router's masking destination addresses, and the destination border router's masking source addresses can help increase sender-receiver unlinkability against observers in the source LAN segment, transit ASes, and observers in the destination LAN segment, respectively, especially when the receiver is a server that does not change its addresses frequently, the destination AS does not upgrade, or the sender uses the same APA to send multiple flows. If a server never changes its address, the sender-receiver unlinkability against the source AS will decrease to zero.

VIII. Related Work

Balancing Accountability and Privacy: Persona [33] balances accountability and anonymity for the next generation Internet (NGI). In Persona, routers encrypt the source IP address of each packet at each hop to hide the source identity of the packet. A victim can ask the last router to backtrace all routers by decrypting the IPs to provide accountability. Although Persona improves anonymity for NGI, it may not provide correct accountability due to route asymmetry and the lack of integrity protection for packets.

The Accountable and Private Internet Protocol (APIP) [7] balances accountability and privacy at the network layer. It splits source addresses into two fields: an accountability address and a return address. Accountability addresses point to *accountability delegates* that vouch for the source's packets. While accountability is handled by delegates, return addresses can be masked to protect the source's privacy. Sources send the fingerprints of packets to their accountability delegates, which work as proof of vouching when verifiers challenge them. However, APIP also has limitations. First, not all packet flows originating from a sender can be shut off because a malicious sender can ignore reporting fingerprints of packets to its delegate once it is verified over a certain time. Second, Design 2 of APIP requires that hosts save the hash of the packets. It is possible for hosts to deny sending specific packets when their accountability delegates verify shutoff requests.

In the Accountable and Private Network Architecture (APNA) [8], ISPs link every packet to its sender without revealing the sender's identity through a cryptographic token called the ephemeral identifier (EphID). Privacy in APNA is not limited to host anonymity but extends to data confidentiality. However, it does not ensure data confidentiality. APNA introduces receive-only EphIDs ($EphID_r$) to prevent the threat of attackers continuously sending shutoff requests against a victim domain. But nodes in the communication path between client and server can launch man-in-the-middle attacks because the client never knows the $EphID_s$ used to establish a connection with the client by the server. Another issue is that large-scale modifications of DNS records enhance the difficulty of deployment of APNA.

Accountability: Some protocols, such as AaaS [34], AIP [35], BAFI [36], and IPA [37] have been proposed to address accountability. AaaS is the first scheme to consider accountability as a service but is costly. AIP and BAFI require host renumbering, and IPA provides prefix-level accountability. None of these protocols addresses privacy.

TABLE VI
DEPLOYMENT COMPARISON WITH OTHER PROTOCOLS.

Protocol	Host	Applications	Access Router	Border Router	Intra-AS Routing	BGP	DNS
APIP [7]	√	√	√	√	√	√	√
APNA [8]	√	√	√	√	√	√	√
PAVI	√	×	×	√	×	×	×

* √ indicates upgrade required. × indicates not required.

Privacy: Some mechanisms have been proposed to support privacy. The IPv6 Internet supports source anonymity to some extent through the use of temporary addresses [38] at the expense of accountability. Moreover, such anonymity is constrained to the day level because a host regenerates its temporary address every other day by default. IPsec [15] is supported naively in IPv6 and maintained by IETF, and can be used to ensure data confidentiality and integrity. Overlay systems such as onion routing [5] and mix networks [6] are designed to provide anonymity at the cost of accountability.

IX. CONCLUSION

In this paper, we proposed PAVI, which bootstraps accountability and privacy in the IPv6 Internet. For the former, it allows user-level associations with packets. For the latter, it achieves sender anonymity, sender-flow unlinkability, sender-receiver unlinkability, and data confidentiality. The results of a quantitative analysis and performance evaluation show that PAVI can achieve accountability and privacy, and is a lightweight and deployable protocol.

X. ACKNOWLEDGMENTS

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